INCH-POUND

MIL-P-28520C(YD)
15 January 1992
SUPERSEDING
MIL-P-28520B(YD)
19 February 1987

MILITARY SPECIFICATION

PUMPING ASSEMBLY, FLAMMABLE LIQUID, BULK TRANSFER, 600 GALLONS PER MINUTE, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, skid-mounted, diesel-engine-driven, centrifugal pumping assembly for use in flammable liquid bulk transfer.

1.2 Scope. Pumping assemblies covered by this specification are of the following classes, as specified (see 6.2).

Class I - 600 gallons per minute (gpm) - JP-5 fuel at 350 feet total dynamic head (TDH).

Class II - 600 gpm - JP-5 fuel at 350 feet TDH.
800 gpm - Nonpotable water at 330 feet TDH.

Class III - 600 gpm - Navy special fuel oil at 330 feet TDH.
900 gpm - Navy special fuel oil at 300 feet TDH.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents .

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-859 - Fuel Oil, Burner.
MIL-T-5624 - Turbine fuel Aviation, Grades JP-4 and JP-5.
MIL-P-16789 - Pumps (Including Prime Movers and Support Items); Packaging of,
MIL-C-27487 - Coupling Halves, Quick-Disconnect, Cam-Locking Type.

STANDARD

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of military and federal specifications and standards are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, CFR, Part 1910.95 - Occupational Noise Exposure.
Title 29, CFR, Part 1910.219 - Mechanical Power-Transmission Apparatus.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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HYDRAULIC INSTITUTE

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Avenue, Cleveland, OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. , 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pumping assembly shall consist of a pump and a diesel engine complete with all necessary accessories for use in flammable liquid bulk transfer.

3.2 First article. When specified in the contract or purchase order, one complete pumping assembly shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term 'recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Selection of material for pump parts. Unless otherwise specified herein, the selection of materials used in the manufacture of the pump shall be in accordance with the Hydraulic Institute Standards. The manufacturer shall select the materials which shall minimize galvanic corrosion to the greatest extent possible.

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3.4 Design. The complete pumping assembly shall be designed to withstand the strains, jars, vibrations, and other conditions incidental to shipping, storage, installation, and operation. The pumping assembly shall be complete so that, when installed, it can be used for the operation for which it was designed.

3.4.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and finished product, provided form, fit, and function requirements are satisfied.

3.5 Performance. The class I pumping assembly shall deliver not less than 600 gpm of JP-5 fuel, MIL-T-5624, against a TDH of 350 feet. The class II pumping assembly, in addition to fulfilling the requirements of class I, shall deliver not less than 800 gpm of nonpotable water, against a TDH of 330 feet. The class III pumping assembly shall deliver not less than 600 gpm of Navy special fuel oil, MIL-F-859, against a TDH of 330 feet. For all classes of pumping assemblies, the maximum net positive suction head requirements shall not exceed 13 feet at the operating speed of the pumping assembly. When tested in accordance with 4.4.2.2, the pumping assembly shall develop a peak efficiency of not less than 60 percent.

3.6 Pump design. The pump shall be centrifugal and self-priming. The pump shall have 6-inch flanged suction and discharge ports. Flanges shall have an equidistant bolt pattern of eight holes, 13/16-inch diameter, on a 9-1/2-inch diameter bolt circle. The pump shall be bronze-fitted, cast iron construction with a self-lubrication mechanical seal. Casing shall be uniform in quality and condition, free from cracks, hard spots, porosity, shrinkage defects, misplaced coring, and other defects which may affect the intended use.

3.6.1 Hydrostatic pressure. The pump shall withstand an internal hydrostatic pressure of 1.5 times the maximum allowable working pressure for not less than 30 minutes without rupture or distortion to any component, and without any leakage through the pump, fittings, piping, or joints.

* 3.6.2 Special accessories. Class I pumping assembly shall be equipped with the following accessories for use with lightweight hose:

a. Two each, flange to quick-disconnect hose coupling halves conforming to MIL-C-27487, one male and one female; circular flange, size 6-inch; aluminum.
b. Suction line strainer; aluminum.
. Suction gate valve; aluminum.
d. Air eliminator; aluminum.

3.7 Diesel engine. The diesel engine shall be an industrial, liquid- or air-cooled, two-stroke-cycle or four-stroke-cycle engine. The continuous horsepower output-shaft rating shall be sufficient to meet the input requirements of the pumping assembly without exceeding the manufacturer's published continuous horsepower. The engine shall be furnished with the following:

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a. 12-volt electric cranking system with a charging system and battery(s) suitable for 20 degrees Fahrenheit (°F) starting.
b. Air cleaner, dry type with replaceable element.
c. Overspeed shutdown control.
d. High temperature shutdown control.
e. High temperature product shutdown control.
f. Engine oil low pressure shutdown control.
g. Spark-arresting muffler.

3.7.1 Class I engine. The class I pump engine shall be equipped with the following special governor control for use with light-weight hose:

a. Speed reducer to no-flow at 120 pounds-force per square inch, gage (psig) discharge pressure.
b. Automatic control to allow not more than 20 inches of mercury vacuum on the suction line.
c. Automatic low discharge pressure shutdown at 80 psig.

When specified (see 6.2), the class I pump engine shall also be equipped with a hydraulic starting system, in lieu of an electrical system, capable of operation in an environment of 20°F.

3.7.2 Manual engine-speed control. The manual engine-speed control shall be the push-pull, locking micrometer-type throttle. The throttle shall operate the engine at any selected speed from idling to maximum governed speed.

3.8 Engine housing. The engine and radiator, if applicable, shall be provided with a sheet metal fabricated housing without side panels.

3.8.1 Instrument panel. The instrument panel shall be flush-mounted on the engine housing. The location shall be such as to permit simultaneous use of instruments, engine controls, and pump discharge controls.

3.9 Fuel system. The fuel system shall have sufficient capacity to provide four-hour operation without refueling. A lever-operated changeover valve having external threads for attachment of a 1/4-inch hose shall provide an outlet connection for an auxiliary fuel system. A 2-1/2 ounce sight-feed oil cup, with a regulating device for adding lube oil when JP-5 is used, shall be mounted within the engine enclosure.

3.10 Grounding assembly.

3.10.1 Ground wire. One uninsulated braided ground wire, 10 American Wire Gage (AWG), 6 feet long, shall be provided. A 50-ampere electrical bronze clamp shall be soldered to each end of the wire.

3.10.2 Ground rod. One copper-clad ground rod, 6 feet long, shall be provided. The rod shall be 1/2-inch diameter, of copper-clad, cold-rolled steel. Suitable clip(s) shall be attached to the skid mount for securing the ground rod to the pumping assembly when the rod is not in use.

3.10.3 Ground terminal. A ground terminal shall be provided on the skid mount. The terminal shall be bonded to the skid mount and shall be located near the engine.

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3.11 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.12 Safety. All mechanical and electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The pump shall comply with Title 29, CFR, Part 1910.95 for noise, and Title 29, CFR, Part 1910.219 for guarding.

* 3.13 Skid mount. The components shall be mounted on a common skid mount. The skid mount shall consist of two steel skid runners with an integrally formed, welded, or bolted base plate for mounting components of the pumping assembly. The skid runners shall be upturned or beveled at each end to permit skidding the pumping assembly into place. The skid runners shall be furnished with mounting holes for anchoring the pumping assemblies to concrete foundation or wooden deck. The number and size of holes shall be as determined by the manufacturer to be required for anchorage of the pumping assembly under the most severe condition of vibration or thrust likely co develop during operation. The skid runners shall be of sufficient height to provide ready access to any connections located under the pumping assemblies and shall be of sufficient width to provide adequate bearing surface on a smooth gravel base. Individual mountings shall be furnished as required, on the base plate for supporting motors , pumps, and other components in vertical alinement. Other components requiring removal for service, adjustment, or horizontal alinement shall be mounted in a manner to permit such removal or alinement. The skid mount shall be provided with fork truck pockets to permit forklift handling. The pockets shall be a minimum of 24 inches on centers and shall be shaped to accommodate a forklift fork of sizes up to three inches in thickness by eight inches in width.

* 3.14 Lifting atttachments. Permanently affixed attachments shall be provided to permit the pumping assembly to be lifted in its normal operating position. The class I pumping assembly shall utilize a single point lifting device; classes II and III may use a single-point lifting device. Each attachment shall withstand, without damage to any part of the assembly, 2.5 times the resultant vectorial force imposed when the individual pumping assembly is suspended by the lifting attachments. Attachments for multiple slings shall be so located that slings will not converge at a point not more than 19 feet above the lowest extremity of the pumping assembly, and not less than l-inch clearance will be maintained between the slings and the pumping assembly. The inside diameter of the attachment eye shall be not less than 3 inches. Lifting attachments may be used for tiedowns, providing they meet the requirements specified in 3.15.

3.15 Tiedown attachments. Permanently affixed attachments shall be provided that permit fastening the pumping assembly to the floor or deck of a transportation medium. Each attachment shall withstand, without damage to the pumping assembly, its proportion of the following static loads: 8G forward, 8G rearward, 2-1/2G upward, 1-1/2G sideward, where G equals the cargo weight. All edges of the attachments shall be rounded or beveled. Tiedowns may be used for lifting providing they meet the requirements specified in 3.14.

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3.16 Inchangeability. All pumping assemblies of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies , accessories, and spare parts.

3.17 Instruction plates. The pumping assembly shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.18 Identification marking. Identification shall be permanently and legibly marked directly on the pumping assembly or on a corrosion-resisting metal plate securely attached to the pumping assembly at the source of manufacture . Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

* 3.19 Cleaning, treatment, and painting. Surface nornally painted in good actice shall be cleaned, treated, and painted as specified herein. Unless otherwise specified (see 6.2), the color of the finish coat shall be green, No. 14064 in accordance with FED-STD-595. Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.20 Workmanship.

3.20.1 Bolted cormections, Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.20.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

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3.20.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.20.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.20.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reservces the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one complete pumping assembly when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items co be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1 and 4.4.4, and the packaging inspection of 4.5.

4.3 Examination. Each pumping assembly shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements , or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection,

4.4 **Tests**.

4.4.1 Hydrostatic test. The pump and fitting shall be subjected to an internal hydrostatic test pressure of 1.5 times the maximum allowable working pressure, for a period of not less than 30 minutes. Nonconformance to 3.6.1 shall constitute failure of this test.

4.4.2 Performance. All data shall be corrected to sea level conditions, barometric pressure 29.92 inches of mercury, for JP-5 and Navy special fuels at 60°F. Water at a temperature to 80°F shall be used as a test fluid.

4.4.2.2 Performance test. The pumping assembly shall be operated continuously using water for a period of not less than one hour. The test shall be conducted in accordance with centrifugal pump requirements of the Hydraulic Institute Standards. During this run, a characteristic head-capacity tune shall be plotted over the complete operating range of the pump while the engine speed controls are set at an adjustment not exceeding that producing the net continuous brake horsepower rating of the engine. Performance at rates less than those specified in 3.5 shall constitute failure of this test. After the one-hour test, the pumping assembly shall be run continuously for not less than 30 minutes through a range of speeds obtained with minimum-to-maximum engine speed control settings. No critical speeds shall be permitted at any point of the range. Each pumping assembly shall be tested for smoothness of operation and sealing effectiveness of mechanical seals. No leakage shall be evident in the casing or through the casing gasket. Minor leakage through the mechanical seals , not to exceed one drop per minute, shall not be cause for rejection.

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4.4.3 Operational test. The pumping assembly shall be operated for 24 hours at the rated conditions specified in 3.5. The pump shall be examined at the end of each 8-hour interval during the 24 hours. Maintenance and minor adjustments shall be permitted during the examination periods. The pump shall be examined during the operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component, or inability of the pump to deliver the minimum capacities specified herein shall constitute failure of this test.

4.4.4 Functional test. The pumping assembly shall be operated for period of not less than one hour to verify the functional operation of the controls. The control functions shall be in accordance with the applicable requirements of 3.7.

4.4.5 Tilted position. The pumping assembly shall be operated for not less than 30 minutes while it is positioned 15 degrees from level along the longitudinal centerline of the skid base. Evidence of malfunction or misalinement of components shall constitute failure of this test.

4.4.6 Cold starting. The pumping assembly shall be placed in a cold chamber at 20°F for 48 hours or until stabilization temperature is reached. The system shall demonstrate three successive starting cycles without the use of external power. Sufficient time should be allowed so that components can return to 20°F equilibrium.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-P-16789. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Presercatuib, packing and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-16789, with the level of presentation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pumping assembly is used primarily to transfer flammable liquids from one storage tank to another. The class II pumping assembly can also be used to pump water for construction, irrigation, or discharge purposes,

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* 6.1.1 Selection factors .

a. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the pumping assembly is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of pumping assembly required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for inspection and approval (see 3.2 and 4.2.1).
e. Male or female quick-disconnect hose coupling (see 3.6.2).
f. When class I pumping assembly is to be equipped with hydraulic starting system (see 3.7.1).
g. When lubrication is to be different (see 3.11).
h. Color of finish coat, if different (see 3.19).
i. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.5).
j. Level of presentation and level of packing required (see 5.1).

6.3 Data requirements . When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one pumping assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Part or identifying number (PIN). The PIN to be used for pumping assemblies applied to this specification is created as follows:

	M28520	X
Specification part number		
Class code (see 6.5.1)		

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* 6.5.1 Class to code identifier. The class of the pumping assembly is identified by a single-digit number as shown in table 1.

TABLE I. Class to code identifier.

Class	Code number
Class I	1
Class II	2
Class III	3

* 6.6 Subject term (key word) listing.

Centrifugal pump
Skid mounted
Navy special fuel oil
JP-5 fuel
Quick-disconnect flange

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity:
Navy - YD

(Project 4320-N340)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-28520C(YD)	920115

3. DOCUMENT TITLE PUMPING ASSEMBLY, FLAMMABLE LIQUID, BULK TRANSFER, 600 GALLONS PER MINUTE, DIESEL-ENGINE-DRIVEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
ANNE L. LYONS	805-982-5761	551-5761
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1564A) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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